

Los Angeles Regional Water Quality Control Board

March 26, 2024

Lysa Gaboudian, Supervising Engineer
County Sanitation Districts of Los Angeles County
Joint Outfall System
1955 Workman Mill Road

APPROVAL OF EPA-821-R-91-100 DRAFT ANALYTICAL METHOD FOR DETERMINATION OF ACID VOLATILE SULFIDE IN SEDIMENT – WASTE DISCHARGE REQUIREMENTS AND NATIONAL POLLUTANT DISCHARGE ELIMINATION SYSTEM (NPDES) PERMIT FOR THE A.K. WARREN WATER RESOURCE FACILITY (ORDER NO. R4-2023-0181, NPDES NO. CA0053813, CI NO. 1758)

Dear Lysa Gaboudian:

Acid volatile sulfide (AVS) is required to be analyzed in receiving water sediments annually, per section 8.3.1.b of the Monitoring and Reporting Program (MRP) of Order No. R4-2023-0181. Footnote b of section 8.3.1.b states that AVS shall be analyzed using the analytical methods appropriate for solid matrices such as ELAP-accredited methods from USEPA SW-846 or other methods approved by the Los Angeles Water Board, State Water Board, or USEPA Region 9.

On December 4, 2023, the Los Angeles Regional Water Quality Control Board (Los Angeles Water Board) received a request from the Los Angeles County Sanitation Districts (LACSD) to use EPA-821-R-91-100 *Draft Analytical Method for Determination of Acid Volatile Sulfide in Sediment* since there is no ELAP-accredited method available for AVS in sediment. On January 8, 2024, the Los Angeles Water Board received a revised request, providing additional rationale for selecting this draft method for AVS sediment monitoring. LACSD provided the following additional information comparing methods EPA-821-R-91-100 to SW-846 9030B for AVS in sediments:

1. AVS is operationally defined as sulfides that form hydrogen sulfide (H_2S) under the conditions described in draft EPA-821-R-91-100 method, i.e. evolution of H_2S using hydrochloric acid (HCl) at room-temperature and releasing the H_2S by bubbling nitrogen or argon into the sample.
2. The SW-846 9030B procedure heats the sample and acid at $100^{\circ}C$ and the H_2S is extracted by distillation in contrast to the room temperature purging used in draft EPA-821-R-91-100 method.
3. The distillation process in SW-846 9030B spans 90 minutes, whereas the purging step in draft EPA-821-R-91-100 method is completed within 60 minutes.

NORMA CAMACHO, CHAIR | SUSANA ARREDONDO, EXECUTIVE OFFICER

4. SW-846 9030B Section 1.2 explicitly states that the method *provides only a semi-quantitative determination of sulfide compounds considered "acid-insoluble" in solid samples*, underscoring the potential inaccuracies in using this method.

Since there is no ELAP-approved method for AVS in sediments and methods included in SW-846 are semi-quantitative, the Los Angeles Water Board finds draft method EPA-821-R-91-100 to be acceptable to use for determination of AVS in sediment. The Los Angeles Water Board hereby approves the use EPA-821-R-91-100 draft analytical method for AVS in sediment for compliance with the monitoring requirements in Order No. R4-2023-0181.

If you have any questions, please contact Dr. Don Tsai at Don.Tsai@waterboards.ca.gov or Steven Webb at Steven.Webb@waterboards.ca.gov.

Sincerely,

Jenny

Newman



Digitally signed by Jenny

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Date: 2024.03.26 20:36:06

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for Susana Arredondo
Executive Officer